Progressive Task-Based Universal Network for Raw Infrared Remote Sensing Imagery Ship Detection

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***Abstract*— Infrared remote sensing images are becoming increasingly popular due to their superior penetration and resistance to light interference. However, challenges still remain when applying them in real-world applications: 1) raw infrared images suffer from severe stripes interference and the preprocessing techniques used to obtain standard image products for subsequent detection tasks tend to be time-consuming, which fails to meet the application requirements; 2) current destriping techniques may inevitably weaken the local contrast between some objects and the local background since they need to consider the gray consistency of the overall image; and 3) in low-resolution images, dim and small infrared targets are challenging to discriminate, resulting in high false alarms. To address these challenges, we proposed a progressive task-based universal network for raw infrared image ship detection while simultaneously removing stripes. First, we built an integrated network consisting of two components: the stripe denoising component (SDC) and the object detection component (ODC). We also designed a feedback loss adjustment mechanism to enhance the focus of the SDC on the target area. Second, a directed two-branch network was constructed for efficient stripe noise removal, including an x-direction branch for feature enhancement and a y-direction branch for grayscale smoothing. Finally, a parallel network with two labels was designed to extract the inherent features of the target and the background, as well as their relationship features, to achieve refined ship detection. We conducted experiments on a self-assembled dataset from the GaoFen-1 satellite to validate our approach. The experimental results demonstrated that the proposed method outperformed other state-of-the-art methods in infrared image ship detection.**

***Index Terms*— Deep learning, infrared remote sensing images, progressive network, ship detection.**

I. Introduction

THE capability of satellite-based target detection, identification, and tracking has steadily improved in both military and civilian applications [1], [2], due to sustained advancements in remote sensing space technology. Compared with other detection systems, satellite-based infrared detectors offer several advantages, such as long-range operation, continuous monitoring, robust anti-interference, and cloud penetration ability. As a result, the study of infrared remote sensing images has unrivaled theoretical value [3].

Due to the voltage difference between the readout circuits of the infrared focal plane array on the satellite sensors, light and dark stripes appear on the generated infrared image. As a result, raw infrared images are often contaminated by stripe noise, severely affecting the visual quality and subsequent applications [4], as shown in Fig. 1(a). Existing infrared image target detection methods have been primarily designed for destriped standard remote sensing data products, which require time-consuming destriped processing [5] and are unsuitable for time-urgent applications. A direct ship target detection method in raw remote sensing infrared images is urgently needed for practical applications. Furthermore, to ensure the gray consistency of the whole image, destriping may reduce the contrast between small targets and their backgrounds, resulting in suboptimal object detection results. Thus, developing an integrated network that can quickly and accurately detect targets based on raw infrared images is a paramount problem to be solved. Although significant progress has been made, destriping and object detection tasks still present challenges, leaving room for improvement. These challenges not only pose difficulties in infrared image target detection research but also offer the possibility of innovative breakthroughs in application scenarios.

At the moment, the challenges of detecting ships in raw infrared remote sensing images are given as follows.

1) Complex stripe noise disturbs the raw infrared remote sensing image, and generating standard image products for object detection is time-consuming, which is difficult to meet the real-time requirements of practical applications.
2) To emphasize the grayscale consistency of the whole image, a separate denoising operation may lead to a reduction in contrast between some dim objects and the local background, which will negatively affect object identification accuracy.
3) For low-resolution remote sensing images, destriping operations may greatly weaken on dim and small objects, and the insufficient features may increase missed detection and false alarms.

Based on the above analysis, developing an efficient universal ship detection approach that can simultaneously perform image destriping and object detection is of paramount research significance and application value.

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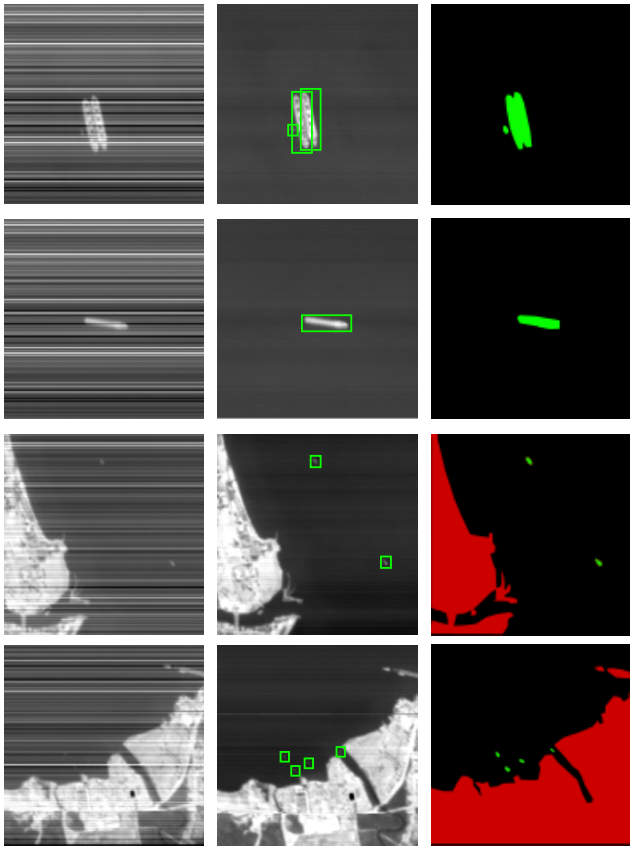

Fig. 1. Schematic of raw infrared remote sensing images and the ground truth of detection. (a) Original infrared image blocks with stripe noise. (b) Ground truth of object boxes. (c) Ground truth of pixel classes.

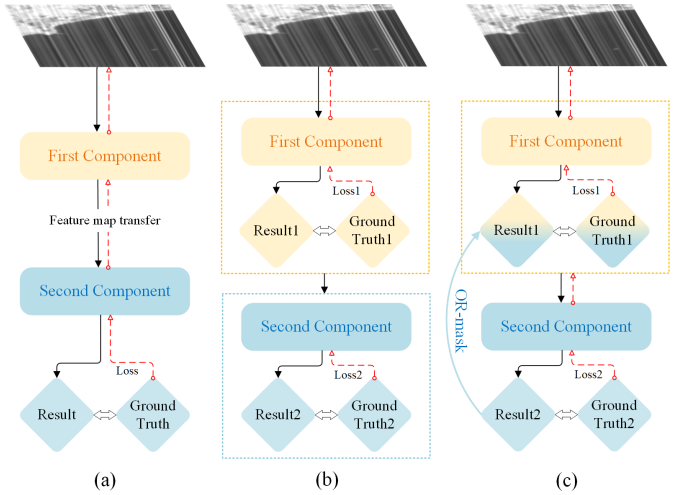

Fig. 2. Illustration of different multicomponent frameworks. (a) Classic multimodule structure. They transfer the intermediate results through feature maps. (b) Common multisupervised architectures where different components are trained separately. (c) Proposed approach uses a universal network to train different components together. OR-mask is a mask of object region.

Previous research efforts on image destriping have included both feature-driven methods [6], [7] and deep learning methods [8], [9]. Feature-driven methods have made significant progress in destriping by designing various handcrafted priors for stripes. However, these methods have required complex models to achieve the desired effect, which are difficult to optimize. In recent years, deep convolutional neural network (CNN) models have shown great potential in destriping due to their powerful feature modeling capabilities. However, these discriminative algorithms have ignored the internal directional features and spatial relationships of stripes, focusing solely on learning from external datasets. Furthermore, there has been a lack of methods that combine destriping and object detection tasks. Typically, destriping is implemented as an upstream task of object detection in multitask cascading methods. They generally consist of two common architectures, as shown in Fig. 2(a) and (b). Specifically, one of them cascades different components through feature maps and the whole network is regulated by a single ground truth [10], [11]. Although they are highly integrated, they are unable to use a divide-and-conquer scheme to reduce the difficulty of complex tasks. Another architecture involves training various components independently under their own supervision before integrating them [12]. However, the learning process of the former component has been supervised only by the manually defined ground truth in this architecture and has ignored the association with the final task goal. It is worth noting that our ultimate learning goal is to achieve better object detection performance rather than simply obtaining clearer images defined by humans.

Based on the above analysis, we proposed a progressive task-based universal network for raw infrared remote sensing imagery ship detection. Specifically, we built a dual-supervised network consisting of a stripe denoising component (SDC) and an object detection component (ODC) for ship detection in low-resolution infrared images. First, the raw infrared remote sensing images were cropped into small patches and fed into the SDC. This component consists of two directed branches for efficient stripe noise removal. The x-direction branch, consisting of a U-shaped network with several gradient filters, was designed for feature enhancement. The y-direction branch was devised for grayscale smoothing and consists of a U-shaped network with several y-directed subspace attention modules. The two feature maps were then fused by a convolution layer to produce the final denoising results. Second, the results from SDC were sent to the ODC, which is a parallel network with two labels (object boxes label and pixel classes label). These two branches were used to extract the different ground features and seek the relationship between the object and their surrounding background features, thereby enabling accurate object detection. Besides, we presented a feedback regulation scheme that enhanced the SDC's focus on objects using an object region mask (OR-mask) derived from the detector. We also constructed a real dataset acquired by the sensors on GaoFen-1 satellite. Comparative experiments demonstrated the superiority of our approach.

The main contributions of this article are listed as follows.

1) A progressive task-based universal framework with two supervised components was developed to process different tasks simultaneously. It can adaptively remove the stripe noise and detect ship, which is in line with the requirements of time-urgent applications.
2) To solve the problems existing in destriping and detection on the raw low-resolution infrared image, we proposed the SDC and ODC separately, both with

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a parallel architecture, to weaken the negative effect of the destriping process on the target features and improve the detection performance.

3) A novel OR-loss regulation strategy was designed to enhance the focus of the SDC on the object region, enabling different components to have accurately detecting the target as the learning objective.
4) An infrared remote sensing image ship detection dataset (named ISD) was built for destriping and ship detection. The detection results for the compared methods verified the practicability of the proposed method.

The rest of this article is arranged in five parts. Section II presents the related work. Section III introduces the model of the proposed universal network. Section IV describes experiments, results, and ablation studies. Finally, a summary is made in the last Section.

II. Related Works

A. Destriping Methods

Typically, there exists nonuniform distribution of reading circuits for infrared detectors. The stripe noises, caused by the nonuniformity in the detector, degrade image quality and may even prevent subsequent image processing, such as image classification, object detection, and recognition [13], [14]. Consequently, it is crucial to find ways to remove stripes while preserving image details. In recent years, numerous methods have been proposed, including filter-based methods, statistical matching-based methods, total variation-based methods, and deep learning-based methods.

Among them, a filter-based method includes the Fourier domain filter [15], [16], wavelet analysis [7], [17], [18], and the combined wavelet Fourier filter [19], [20]. Since the periodic nature of fringe noise is easy to be identified in the power spectrum, filter-based methods perform well in dealing with periodic noise. Another class of popular denoising methods is the statistical matching-based method, which was originally applied to marine data. These approaches can be classified as equalization methods [21], histogram modification [22], and moment matching [23]. Scholars have also combined local constant statistical constraints with neural networks to remove the stripe noises [24]. However, these methods are efficient only for specific distributions of stripes. In addition, total variation-based minimization methods have also emerged as a class of denoising methods. They transform the denoising task into an optimization problem and rely on gradient descent flow to smooth the image [25], [26], [27]. In [25], a novel variational model considering different directions was proposed to achieve better destriping results. Then, the L1-norm was treated as the loss function to enhance the denoising performance [28]. Wang et al. [29] proposed a new total variation coupled with a guided filter (TVGF) for destriping, which can also improve the denoising performance. Compared with other methods, the total variation-based methods give more evenly balanced results both quantitatively and qualitatively. However, the resulting images may be oversmoothed in complex scenes.

Moreover, modern deep learning approaches are emerging, and making the problems in this field can be solved smoothly [30], [31], [32]. Kuang et al. [30] first introduced the SNRCNN, a deep learning method for removing the infrared image stripes. This was a beneficial attempt, but its effectiveness was limited by network capacity. In [31], a feed-forward denoising convolutional neural network (DnCNN) was designed, which incorporated many strategies into image denoising. Similarly, in [33], a fast and flexible network (FFDNet) was presented to handle spatially variant noise. This network treated a tunable noise image as input and achieved good denoising performance. Guan et al. [34] proposed a two-stage cascaded residual CNN to reduce noise, which enhanced the model performance by combining multiple convolutions. Although all of these methods can produce effective denoising results, they disregard the goals of subsequent tasks. Using undifferentiated denoising methods can inevitably affect the accuracy of downstream tasks to some extent. Therefore, it is important and worthwhile to consider combining denoising with other tasks.

B. Ship Objects Detection Methods

Ship object detection on remote sensing images is an essential topic in both military and civilian fields, and the recent success of deep learning approaches has made it a focus of study [35], [36], [37]. However, since infrared remote sensing images usually have low spatial resolution, most existing methods mainly focus on solving the problems of small object scale and scale imbalance. For example, Liu et al. [38] presented a network capable of extracting the feature maps from the layers at different depths, which exhibited strong robustness and better accuracy. In 2019, a dense attention pyramid network [39] was proposed to accurately detect small ships in synthetic aperture radar images. This network utilized densely connected convolutional block attention modules across the entire pyramid structure to extract abundant features and enable refined detection. In addition, Chen et al. [40] proposed a hybrid deep learning method that used generative adversarial network (GAN) and CNN to generate enough small ship samples to improve the robustness of the model. Zhang et al. [41] presented a balance learning network to solve the problems of image sample scene imbalance and ship scale feature imbalance. In summary, the aforementioned methods have improved detector performance to some extent; however, they have not considered the impact of false alarms on detection, which is also a crucial issue.

In addition, it has been observed that ships often have a strong relationship with the sea background, which has been explored by many researchers in recent years [42], [43]. First, according to [44], object–scene relationships have a significant impact on detection performance. In [45], an improved salient object detection was presented for brain image analysis by utilizing both background cues and foreground connectivity. Yin et al. [46] developed a new method for detecting adversarial attacks by capturing the object co-occurrence relationships in complex scenes, achieving better performance than state-of-the-art methods. This is an excellent example of utilizing the relationship between objects and their background or between different objects. These methods demonstrate the benefits of incorporating the relationship between different objects to assist in the target task. Therefore, they have served

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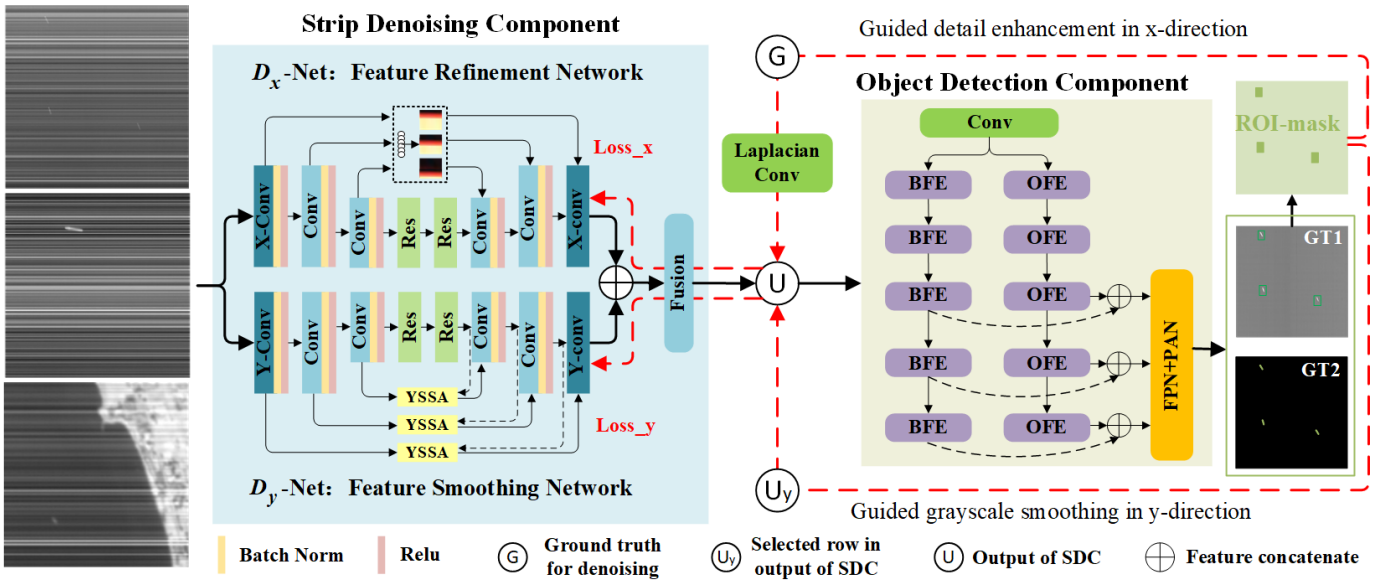

Fig. 3. Pipeline of our progressive task-based network. The proposed framework consists of SDC and ODC.

as a valuable source of inspiration for us to combine the relationship between the ship and the ocean background for accurate object detection.

III. Proposed Method

The overall framework of the proposed network is shown in Fig. 3. It comprises two supervised components, including SDC for destriping and ODC for detection. In practical applications, detecting ship targets in raw level-0 infrared images with stripe noise directly is essential due to the real-time requirement. The SDC, which consists of two direction branches, was used to automatically remove the stripe noises. The ODC was utilized to learn more accurate object representation for infrared remote sensing images. More importantly, the object boxes obtained from ODC were treated as weights mask for SDC to adjust the attention. This scheme enhances integration between different components and directs the network's focus toward ship objects, leading to a more robust detection effect. In the following, we will provide a more detailed explanation of the proposed algorithm.

A. Structure of SDC Component

After conducting a lot of experiments and observations, we have discovered that the directional characteristic of stripe noise plays a crucial role in determining the effectiveness of denoising. Therefore, we constructed a two-branch network for stripe removal. This network comprises a feature refinement branch in the X-direction (D_x-Net) and a feature smoothing branch in the Y-direction (D_y-Net), both of which are U-shaped network architecture with several encoder layers and decoder layers. They utilize the strip kernels (1×5 or 5×1) in the x- and y-directions for the first and last layers. In addition, each network includes two residual modules in its middle layer.

In addition, we introduced a graduated filter group within the skip connections, which is used to adaptively enhance

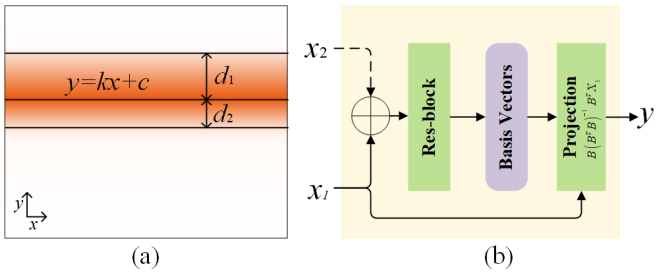

Fig. 4. Schematic of (a) graduated filter and (b) YSSA.

region features in the x-direction. The stripe graduated filter, shown in Fig. 4(a), is defined by three parallel lines. The filter's location and orientation were determined by the central line, which is a standard linear function ($y = kx + c$ with slope k and intercept c). The color in the central line area is deepest, indicating that the pixels in this area are enhanced by 100% or directly multiplied by the enhancement factor e_g. Then, on both sides of the central line, the applied enhanced factor linearly decreases from e_g to 0 (white area). The graduated filter is mathematically described by the following equations:

$$a(x, y) = e_g * \max\left\{0, \left(1 - \frac{s(x, y)}{d_1}\right)\right\} \tag{1}$$

$$b(x, y) = e_g * \max\left\{0, \left(1 + \frac{s(x, y)}{d_2}\right)\right\} \tag{2}$$

where $s(x, y) = y - (kx + c)$ denotes a function of a point (x, y) relative to the central line and d_1 and d_2 are the gradient region distance parameters. Since the features in the x-direction is enhanced, the slope k is set to 0 in this work. c, d_1, d_2, and e_g denote the learnable parameters of this filter. In this block, we use three gradient filters to process the input feature and add the resulting feature maps as the final output.

After that, we developed a network that smooths the feature in the y-direction by introducing a subspace attention module. The process is shown in Fig. 4(b), where x_1 and x_2 represent two feature maps from a single image, obtained from the encoding and decoding layers, respectively. Then, these feature

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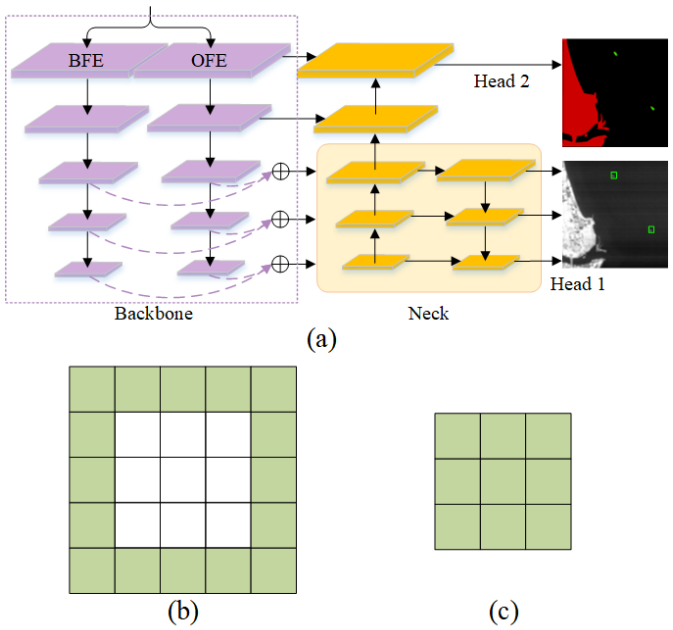

Fig. 5. (a) Schematic of the ODC with two labels. (b) and (c) Convolution kernels in BFE module and OFE module, respectively.

maps need to be concatenated and fed into a stripe res-block module, which includes a stripe convolution kernel, to generate basis vectors $[B_1, B_2, \ldots, B_k]$ for different subspaces. This basis generation operator is given by the expression

$$B = f(x_1, x_2) \tag{3}$$

where B denotes a matrix composed of basis vectors $[B_1, B_2, \ldots, B_k]$. The input feature map x_1 can be projected onto a k-dimensional subspace by orthogonal linear projection. The resulting matrix R is calculated as follows:

$$R = B\left(B^T B\right)^{-1} B^T. \tag{4}$$

The feature map x_1 is reconstructed as y

$$y = Rx_1. \tag{5}$$

Based on this approach, the stripes and features of the original image can be more effectively separated, leading to better noise elimination.

To generate the final denoised output map, the features obtained from the two branches need to be concatenated and calculated by a fusion module. It is worth noting that rather than simply obtaining clear denoised images, the ultimate goal of introducing the SDC is to improve the detection accuracy by providing higher quality input to the ODC.

B. Structure of ODC Component

The ODC component was constructed upon an optimized one-stage detection framework, such as YOLOv4 [47]. As shown in Fig. 5(a), the backbone is a two-stream model with five object feature extraction (OFE) modules and five background feature extraction (BFE) modules. Each feature extraction module includes n residual units, where OFE residual units use standard convolution kernels and BFE residual units use dilated convolution kernels. In this work, the n value for each module layer is set to [1, 2, 4, 4, 2]. After each OFE module, there is a CBL block, which includes a convolution layer, a batch normalization layer, and a rectified linear unit (ReLU) activation layer. Standard convolution kernels are used to extract the object features of the image, while the dilated convolution kernel is designed to capture the surrounding background information, as shown in Fig. 5(b) and (c). Then, in the last three BFE and OFE modules, the output feature maps of the same level are concatenated together as the input to the next neck modules. This process is expressed mathematically as follows:

$$\Psi_{O/B}(I) = \prod_{i=1}^{k} \left(M^i_{\mathrm{CBL}}\left(M^i_{O/B}(I)\right)\right) \tag{6}$$

$$M_{\mathrm{CBL}}(I) = \mathrm{Relu}(\mathrm{BN}(C(I))) \tag{7}$$

$$M_{O/B}(I) = \prod_{i=1}^{n} R^i_{O/B}(I) \tag{8}$$

$$\Psi_{\mathrm{ODC1}}(I) = \mathrm{Fu}\left(\mathrm{Cat}^5_{k=3}(\Psi_O(I), \Psi_B(I))\right) \tag{9}$$

where I represents the input data and $\Psi_{O/B}$ stands for the OFE or BFE network branch. M_{CBL} and $M_{O/B}$ represent the calculation steps of the CBL module and OFE/yBFE block, respectively, and $R_{O/B}$ denotes the residual blocks in the OFE or BFE module. F_u represents the processing operator of the neck module, Cat denotes the concatenate operator, and Ψ_X denotes the calculation process of the X component or module. In addition, $\prod_{i=1}^{n} f_i(I)$ is defined as

$$\prod_{i=1}^{n} f_i(I) = f_n(f_{n-1}, \ldots, f_2(f_1(I))) \tag{10}$$

where f denotes a function or a module that needs to be implemented repeatedly, such as M^i_{CBL} and $R^i_{O/B}$. n stands for the module number in the network. Based on the above processing, the detection results can be obtained.

In addition, previous studies have shown that different task branches could mutually reinforce each other in multitask learning, which could effectively improve the network performance [48]. Inspired by this, we added a segmentation branch to the original detection network to help it to discover the relationship between the objects and backgrounds. The whole network shares one encoder and combines two decoders to complete different tasks. The encoder network is also supervised by the segmentation label, which enables it to understand the class information of ship targets, ocean, land, and their spatial relationship. As a result, the relationship information between objects and backgrounds learned by the network from the segmentation head can assist the detection head in accurately detecting the target and reducing false alarms on land.

It is noteworthy that the proposed method aims to achieve accurate object detection, and then, the segmentation branch is only used during the training phase and not during the test evaluation process. As shown in Fig. 5(a), the results obtained from the neck module are fed into the semantic segmentation branch. By upsampling the features from the neck module and the coding layer of the same level, feature maps with higher

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resolution are obtained. The final feature map has the same size as the original image and represents the probability of each pixel being a ship object, land, or sea background. The processing can be defined as follows:

$$\Psi_{\mathrm{ODC2}}(I) = \prod_{k=2}^{1}\left(\mathrm{Conv}\left(\mathrm{Cat}\left(\mathrm{up}\left(I_k^d\right), I_k^e\right)\right)\right) \tag{11}$$

where Conv and up represent the convolution and upsampling operators, respectively, I_k^d and I_k^e denote the input feature maps from the previous decoding level and the encoding layer of the same level, respectively, and k denotes the sequence number of feature layer.

C. Whole Framework of Our Approach

The whole framework of the developed approach is a dual-supervised framework with two components. As shown in Fig. 3, the input images are computed separately by two components connected sequentially, which is expressed as

$$O = \Psi_{\mathrm{ODC}}(\Psi_{\mathrm{SDC}}(I)) \tag{12}$$

where I and O denote the input and output images, respectively, and SDC and ODC denote two separate models that are combined into a unified architecture for training and testing. It is important to note that each model has its own ground truth that is used for supervision during the training process.

First, a pixel-based loss function, mean absolute error (MAE), is utilized to calculate the training loss of SDC component. Although it is sample, it had been proven to be effective [49]. In this work, we refer to it as the data fidelity term, which is expressed as

$$L_{\mathrm{df}} = \frac{1}{n}\sum_{i=1}^{n}|y_{\mathrm{gt}}^i - y_{\mathrm{pre}}^i| \tag{13}$$

where y_{gt}^i and y_{pre}^i represent the ith pixel in the ground truth and predicted results, respectively, and n represents the number of pixels.

Since the ultimate goal of the proposed approach is to achieve accurate object detection, the SDC also needs to be trained with the purpose of optimizing the detection performance. Based on this consideration, we designed two additional loss functions for supervised model training of SDC: a detail enhancement loss L_{de} and a grayscale smoothing loss L_{gs}. They correspond to Loss_x and Loss_y in Fig. 3, respectively. In specific, the object boxes obtained from ODC are transferred as OR masks to adjust the attention region of SDC. On the one hand, the detail features are obtained by applying Laplacian convolution to the data within these masks. On the other hand, the L_{de} loss is calculated by comparing the prediction result with the ground truth. The loss function is defined as follows:

$$L_{\mathrm{de}} = \frac{1}{n}\sum_{\Omega}\left|\mathrm{LC}\left(y_{\mathrm{gt}}\right)^i - \mathrm{LC}\left(y_{\mathrm{pre}}\right)^i\right| \tag{14}$$

where Ω denotes the OR-mask area, i represents the pixel in the mask region, and LC stands for the Laplace convolution operator.

Moreover, we extracted the pixels of the rows corresponding to the target box and calculated the variation difference in the y-direction as the loss function L_{gs}

$$L_{\mathrm{gs}} = \nabla_y\left(\frac{\nabla_y u}{\sqrt{\left(\nabla_y u\right)^2 + \varepsilon^2}}\right) \tag{15}$$

$$\left(\nabla_y u\right)_{i,j} = \begin{cases} u_{i,j+1} - u_{i,j}, & \text{if } j < N \\ 0, & \text{if } j = N \end{cases} \tag{16}$$

where u represents the predicted result, ε represents a constant, and N denotes the number of rows. Therefore, the total loss of the SDC component is defined as follows:

$$L_S = \lambda_1 \cdot L_{\mathrm{df}} + \lambda_2 \cdot L_{\mathrm{de}} + \lambda_3 \cdot L_{\mathrm{gs}} \tag{17}$$

where λ_1, λ_2, and λ_3 denote constants. The best results were obtained when they are the same according to the ablation experiment in this work, so they were set to 1.

Recently, focal loss [50] has stood out among many methods in the object detection field and can effectively reduce the negative effect of sample imbalance. We, therefore, chose to use it as the classification loss (L_{c1}). Furthermore, L_{c1} together with confidence loss (L_{c2}) and intersection over union loss (L_i) formed the final loss function, which is shown in (18). The specific details of the above loss can be found in [47]

$$L_{\mathrm{det}} = \gamma_1 \cdot L_{c1} + \gamma_2 \cdot L_i + \gamma_3 \cdot L_{c2} \tag{18}$$

where γ_1, γ_2, and γ_3 represent the weight factors, and they were set to 1 relying on the practical experience of previous researcher [47]. Moreover, the segmentation branch employs the cross-entropy loss function, which computes the pixelwise error between the output and ground truth. It is defined as follows:

$$L_{\mathrm{seg}} = -\sum_{i=0}^{K-1} y_i \log\left(\hat{y}_i\right). \tag{19}$$

In conclusion, the final loss function of the ODC component is a sum of two parts

$$L_O = L_{\mathrm{det}} + L_{\mathrm{seg}}. \tag{20}$$

IV. Experiments

In this section, we provide a brief overview of the new ISD dataset for infrared ship detection, followed by a description of experimental setup and evaluation metrics. We then resent the results of our ablation and comparison experiments, where we compare the proposed method with some representative methods to verify its effectiveness. Finally, we analyze and discuss our findings in detail.

A. Experimental Description

1) ISD Dataset: We conducted experiments using the infrared remote sensing images from the GaoFen-1 satellite to evaluate the model performance. The new dataset (called the ISD dataset) includes the images captured in different places and environments. Raw infrared image and the level-2 denoising image product are shown in Fig. 6. The spectrum of

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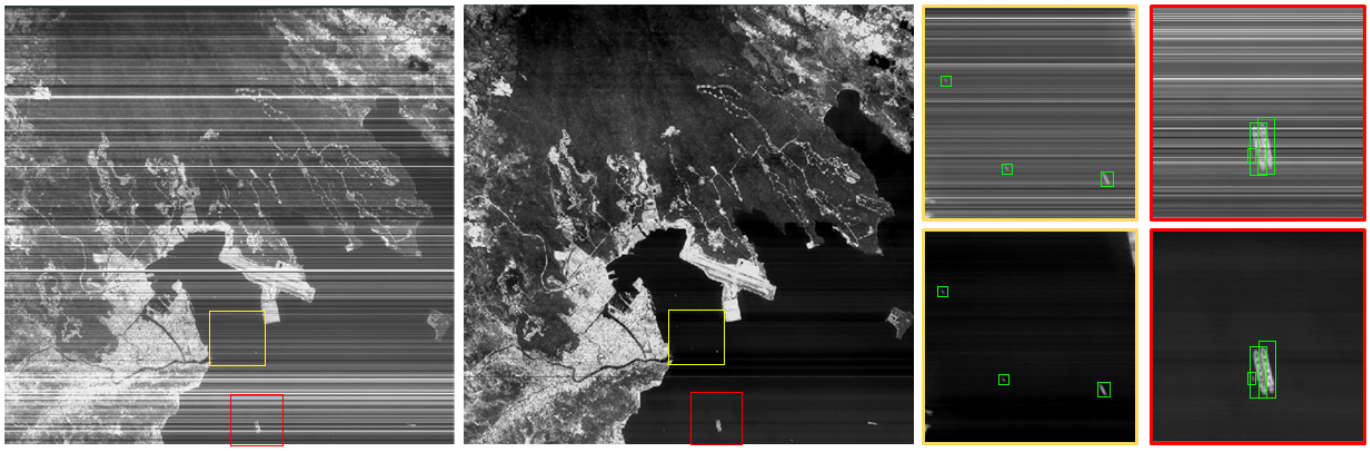

Fig. 6. Schematic of raw infrared image and the denoising image. Green boxes denote the ship targets.

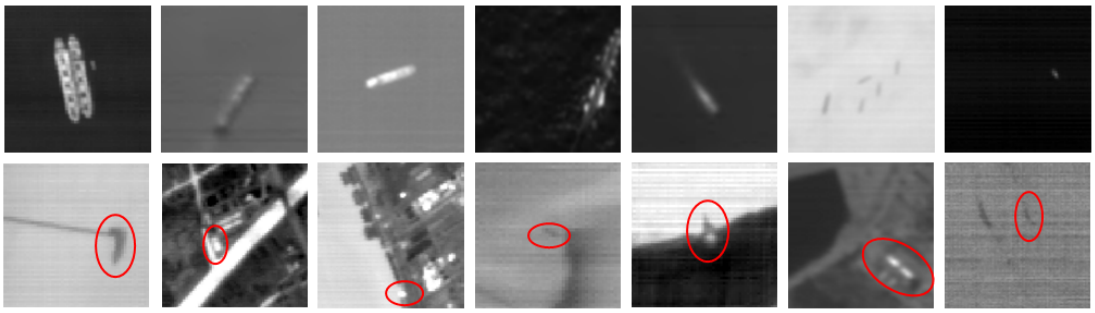

Fig. 7. Images the in first row are the true ship objects, while the images in the second row are the false alarms.

these images varies from 0.77 to 0.89 μm, and the spatial resolution is 16 m. The size of raw infrared remote sensing image product is 10 786 $\times$ 10 080. Because of the large image size, we cut it into patches of 256 $\times$ 256 pixels to meet the training requirements. The final dataset contains a total of 2470 images with 2389 ship instances, and one-third of them are positive samples. We choose 1729 images for training, 247 images for validation, and remaining 494 images for testing.

Due to the characteristics of infrared remote sensing images, object detection based on the ISD dataset faces some challenges. First, there is a significant difference in scale between the targets, which can be seen in Fig. 7. The maximum and minimum target size difference is about a factor of 10, and the targets lack texture information. Second, the target has different gray levels and exhibits a polarity feature. For example, the gray value of target is opposite in day and night images, as shown in the first row of Fig. 7. Third, complex background leads to a higher rate of false alarms, which negatively affects the detection accuracy. For example, the trestle, bright buildings along the river, and streaked clouds in the second row of Fig. 7 greatly interfere with the performance of the model. Furthermore, because our approach was proposed for a progressive task, we designed a cascade network with two levels of supervision. Consequently, not only object boxes but also the obtained level-2 products containing clear images are used as the ground truths.

2) Experimental Details: Our approach was implemented using the PyTorch framework and executed on an RTX 3090 24-GB GPU under Ubuntu 18.04 LTS. To ensure a fair comparison with other methods, we optimized each method as much as possible. It takes 500 epochs for the network to complete training. In our experiments, the intersection over union (IoU), confidence, and NMS thresholds were set to 0.2, 0.3, and 0.5, respectively, to achieve better results for small-sized targets. During the training process, the dynamic learning rate adjustment strategies of SDC and ODC were used with a cosine annealing scheme. The learning rates of the SDC and ODC ranged from 1×10^{-4} to 1×10^{-8} and from 1×10^{-3} to 1×10^{-6}, respectively.

3) Evaluation Index: We picked accuracy evaluation indexes from both the remote sensing processing field and deep learning field to quantitatively assess the ship detection performance of the comparison methods. First, the IoU was adopted as a measure of the ratio between the intersection and union of the predicted rectangular and the ground-truth rectangular. These indicators (detection rate P_d and false alarm rate P_f) are expressed as

$$P_d = N_{tp}/N_{\text{gt}} \tag{21}$$

$$P_f = N_{fp}/\left(N_{tp} + N_{fp}\right). \tag{22}$$

Also, the precision (P) and recall (R) can be calculated as

$$P = N_{tp}/\left(N_{tp} + N_{fp}\right) \tag{23}$$

$$R = N_{tp}/\left(N_{tp} + N_{fn}\right) \tag{24}$$

where N_x denotes the number of $x \in$ [TPs, FPs, FNs, GTs]. They denote the true positives, false positives, false negatives,

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TABLE I
COMPARATIVE EXPERIMENTAL RESULTS FROM DIFFERENT PROGRESSIVE FRAMEWORK

Methods	R	P	AP	Time
A[1]	89.36	85.03	84.08	**30.77**
B[2]	92.72	87.48	87.17	42.51
C[3]	**94.65**	**91.26**	**90.33**	30.81

[1]Classic cascade architecture. [2]Separated dual-supervised architecture. [3]Universal dual-supervised architecture.

and ground truth. For example, the term "TPs" refers to samples whose IoU between the predict box and ground-truth box is greater than the setting threshold. Based on the above description, we can see that since $N_{\text{gt}} = N_{tp} + N_{fn}$, $R = P_d$ and $P = 1 - P_f$. Consequently, the two types of evaluation indicators are essentially the same. In addition, we used the average precision (AP) score to comprehensively evaluate the pros and cons of the detection model. It is obtained by a series of recall and precision value at different thresholds. It can more accurately reflect the advantages and disadvantages of different algorithms. We also used the frames per second (FPS) to show the running speed of each algorithm, which is the inverse of time.

B. Evaluation of the Proposed Scheme

1) Analysis of Progressive Framework: In this section, we conducted a comparative analysis of different types of cascade frameworks, as shown in Fig. 2. The first framework (A) follows the classic cascade architecture, where the SDC and ODC components were connected through feature maps, that is, they were fused into a unified network with only one supervision. The second framework (B) corresponds to a dual-supervised approach with separated components. We trained SDC and ODC separately and associated them during the test phase. The output image of SDC was fed into ODC to obtain the final detection result. The last framework (C) represents the proposed framework, which uses dual-supervised joint mode for different components. The SDC and ODC were supervised by their own ground truth separately, but they were trained uniformly. The association of the two components was achieved through a feedback regulation strategy of an OR-loss. The specific experimental results are shown in Table I.

Compared with the other two methods, the first method (A) performs the worst in terms of AP. This is likely due to the lack of supervision at the first level, which causes the goal value of a task to be significantly different from the initial value. Consequently, the final detection performance is not satisfactory. Method (B) treats image denoising and object detection as separate tasks, which is consistent with current practical applications. Although it improves the detection performance compared to method (A), it takes more time to transfer and load the image during the association of the two tasks. This is fatal for the application with high real-time requirements. Therefore, a progressive task-based universal network, method (C) in Table I, was proposed to detect the ship object on raw infrared images at level-0. This method can achieve accurate object detection directly on the original stripe image without any intermediate processing steps. The experimental results verified the effectiveness of the proposed OR-loss strategy and the overall network architecture design.

TABLE II
EVALUATION OF DIFFERENT METHODS BASED ON THE ISD DATASET

Model	Evaluation indexes			Inference Speed	
	$R(P_d)$	$P(1\text{-}P_f)$	AP	Time	FPS
YOLOv5	85.29	81.42	81.30	**23.31**	**42.90**
ODC*	86.42	83.51	83.29	25.77	38.80
ODC	88.17	85.71	85.39	26.34	37.97
SDC*+ODC	93.75	89.34	89.01	29.76	33.60
SDC+ODC	**94.65**	**91.26**	**90.33**	30.81	32.47

Note: The units of Pd, Pf, R, and P are percentages (%), and the unit of time is ms. * represents the original single-branch architecture without optimization.

2) Analysis of Different Module: In this section, we conducted experiments and analyzed the performance of different modules based on the ISD dataset, including the parallel network with two labels for ODC, a directed two-branch network for SDC, and the dual-supervised universal framework. YOLOv5 was treated as the baseline method due to its superior detection performance. The specific results are shown in Table II. "ODC*" and "SDC*" represent the architecture of their corresponding single-branch model. In specific, "ODC*" represents the method that trains the network with two classes of labels but only uses the single-branch network structure with the standard convolution kernel. "SDC*" denotes a single U-shaped network architecture and does not contain the developed gradient filter and y-directional subspace attention (YSSA) module. All kernels are standard square kernels (such as 3×3) without direction. All the methods used for comparison were conducted based on raw infrared images with stripe noise.

Obviously, we found that the AP of baseline method is lower than that of the "ODC*" method (row #1 versus row #2). This shows that the correlation information between object and background is helpful to achieve accurate detection. The third experiment (ODC) illustrates the complete ODC component with two parallel branches, which leads to a reduced false alarm rate compared with the previous method. Row #4 shows the universal network with "SDC*" and the ODC components, which are cascaded by the OR-loss scheme. It can be observed that this approach produces a competitive improvement over the detection rate, and the overall evaluation index AP has been greatly improved. It is interesting to note that the presented two-branch model for SDC is effective (row #5 versus row #4). Moreover, the feature maps of each method are shown in Fig. 8, from which the advantages of the proposed module can be intuitively seen. Nonetheless, compared with the object detection model alone, there is a slight increase in the testing time. However, it is greatly reduced for cascade tasks with denoising and detection.

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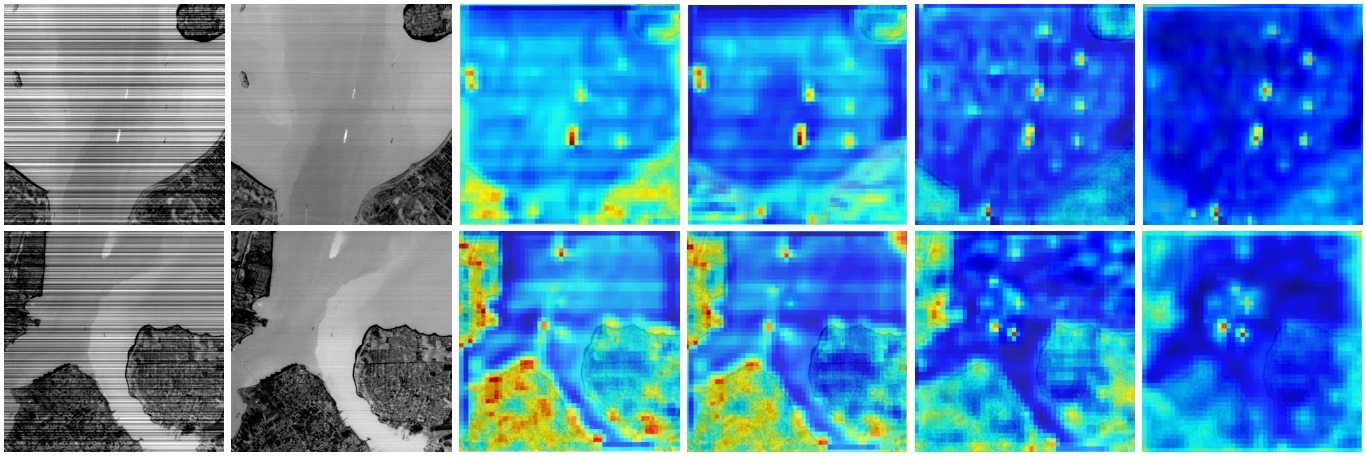

(a) (b) (c) (d) (e) (f)

Fig. 8. Comparison of the different modules. (a) and (b) Infrared images and denoising images, respectively. (c)–(f) Feature maps of ODC*, ODC, SDC* + ODC, and SDC + ODC, respectively.

TABLE III
EXPERIMENTAL RESULTS OF HYPERPARAMETERS λ_n ON THE ISD DATASET

Hyper-paramers			AP
$\lambda_1 = 0.2$	$\lambda_2 = 0.4$	$\lambda_3 = 0.4$	87.45
$\lambda_1 = 0.4$	$\lambda_2 = 0.3$	$\lambda_3 = 0.3$	88.97
$\lambda_1 = 0.6$	$\lambda_2 = 0.2$	$\lambda_3 = 0.2$	87.72
$\lambda_1 = 0.8$	$\lambda_2 = 0.1$	$\lambda_3 = 0.1$	86.17
$\lambda_1 = 1$	$\lambda_2 = 1$	$\lambda_3 = 1$	**90.33**

3) Analysis of the OR-Loss Strategy: Different hyperparameters have a great impact on the model's performance. In this section, we performed the ablation experiments to determine the best value λ_1, λ_2, and λ_3 in the SDC loss function of the proposed framework. They represent the weight relationship between different loss subterms and also illustrate the importance of OR-mask in associating two components. It can be seen in Table III that we adjusted the weight ratio between the data fidelity term and the last two terms from small to large to find the best parameters. Based on the previous experimental experience, the branch proportion in the x- and y-directions should be the same, so λ_2 and λ_3 were set to the same value. The best performance was achieved when all λ_n were set to 1. This indicates that SDC should focus more on the target features, which can improve the final detection performance. Nevertheless, the AP value dropped when the ratio of λ was balanced. This indicates that the overall context is just as important, and it still affects the final detection performance.

C. Approach Comparison and Analysis

To verify the algorithm's effectiveness, we compared the proposed method with other investigated methods based on the ISD dataset. We categorized the comparison methods into three groups. First, four classical methods are included in the comparison methods, namely, Faster-RCNN [51], SSD [52], YOLOv5, and an improved method FMSSD [53] designed for remote sensing object detection. Second, the approaches specifically designed for small object detection were compared, such as R^3-Net [54], MSCNN [55], and SME-Net [56]. Finally, we added other excellent methods to further enrich experiments, such as the classic cascade model CC-Net [10] and an algorithm designed for ship detection, R-DFPN [57]. To ensure a fair comparison, all methods were set to the same hyperparameters, except for the parameters specific to their different modules. We utilized open-source code for all methods during the testing phase.

TABLE IV
PERFORMANCE EVALUATION OF EACH METHOD ON THE ISD DATASET WITH RAW INFRARED IMAGES

Model	Evaluation indexes			Inference Speed	
	R(P_d)	P(1-P_f)	AP	Time	FPS
FasterRCNN	80.11	76.83	76.49	37.84	26.43
SSD	82.85	77.93	77.04	**23.09**	**43.31**
FMSSD	83.58	79.82	78.95	35.54	28.14
R-DFPN	84.14	79.77	79.29	38.09	26.25
YOLOv5	85.29	81.42	81.30	23.31	42.90
R^3-Net	86.40	83.98	83.13	34.97	28.60
MSCNN	85.81	82.74	82.02	31.38	31.88
SME-Net	87.27	84.93	84.09	28.72	34.82
CC-Net	89.62	85.51	85.27	35.01	28.56
Proposed	**94.65**	**91.26**	**90.33**	30.81	32.47

Note: The units of Pd,Pf, R, and P are percentages (%), and the unit of time is ms.

On the one hand, we compared all investigated methods based on the raw infrared images with stripe noise. All models were trained on these images to align with the actual application requirements. We can see the quantitative evaluation results of the model detection performance in Table IV. The seventh column lists the AP values of all methods. Obviously, the proposed algorithm reached the best

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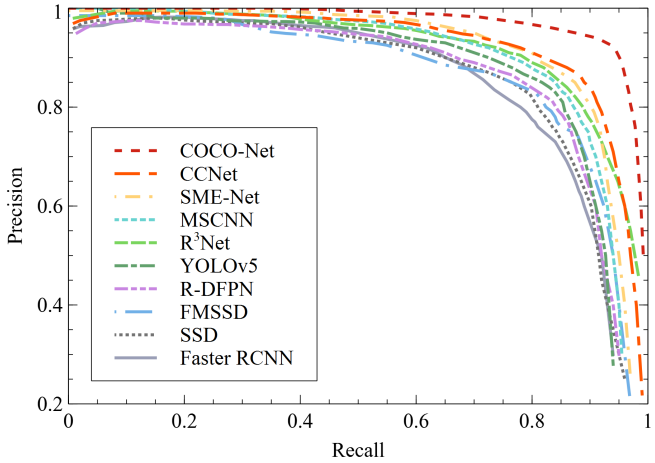

Fig. 9. Schematic of the PR curves. They were obtained by all compared approaches on the ISD dataset with raw infrared images.

performance (bold font), with improved recall and precision rates compared to other methods. In comparison with the multistage framework of CC-Net, the proposed approach, integrating the dual-level supervision with OR-loss feedback control technology, achieved better results. Furthermore, our approach exhibited strong performance in indicators of remote sensing, such as P_d and P_e. Since the images contain many small-sized objects that are difficult to detect due to their lower resolution, we compared the proposed approach with the best-performing small object detection methods, such as SME-Net, MSCNN, and R^3-Net. The results show that AP can be increased by up to 8% at most. In addition, even if the detection time of the proposed method is not the shortest for single detection, it includes a noise removal operator and greatly improved the detection performance. At the same time, the time required for detection method with denoising operator, as mentioned in Section IV-B1, has been effectively shortened.

On the other hand, our approach was also compared with the models trained on clear infrared images to ensure fairness to other models without denoising processing. However, since our method includes an SDC component, the model evaluation was still based on the raw infrared image with stripe noises. The evaluation results can be found in Table V. Compared with the previous experimental results, the detection accuracy of each model on clear images was effectively improved, indicating that image clarity has a significant impact on detection results. It is worth noting that our method outperforms other methods in terms of detection performance, despite processing noisy images. This further validates that the method is well adapted to the task of ship detection in infrared images and can achieve robust performance in practical applications.

Besides, the precision and recall (PR) curves of each model for the above two experiments are shown in Figs. 9 and 10. The PR curve provides a comprehensive evaluation of the detection model by considering both recall rate and precision. Usually, high precision and high recall cannot be simultaneously achieved, which makes it challenging to evaluate a model solely based on one of the metrics. In this case, the PR curve plays a good role and helps to visualize the tradeoff

TABLE V
PERFORMANCE EVALUATION OF EACH METHOD ON THE ISD DATASET WITH CLEAR INFRARED IMAGES

Model	Evaluation indexes			Inference Speed	
	R(P_d)	P(1-P_f)	AP	Time	FPS
FasterRCNN	83.32	79.99	79.33	37.95	26.35
SSD	85.33	80.55	80.17	**22.97**	**43.54**
FMSSD	85.89	81.57	80.66	35.71	28.00
R-DFPN	86.59	84.05	82.03	38.43	26.02
YOLOv5	87.95	83.81	84.02	23.32	42.88
R^3-Net	88.34	85.17	85.27	34.47	28.01
MSCNN	87.49	85.31	84.93	31.29	31.96
SME-Net	90.02	86.48	86.21	29.02	34.46
CC-Net	92.37	88.42	88.95	34.83	28.71
Proposed	**94.65**	**91.26**	**90.33**	30.81	32.47

Note: The units of Pd, Pf, R, and P are percentages (%), and the unit of time is ms.

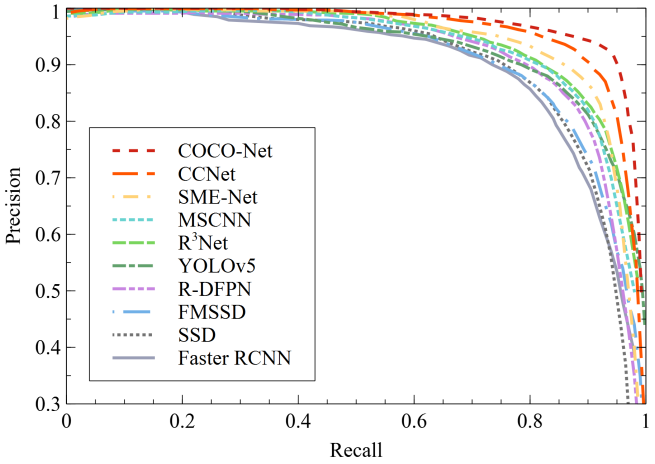

Fig. 10. Schematic of the PR curves. They were obtained by all compared approaches on the ISD dataset with clear infrared images.

between PR at various detection thresholds. For the curve that covers the full range of the horizontal axis, the higher up the curve the better the performance. The PR curves clearly demonstrate the performance differences between the different models.

Observing Fig. 11, it illustrates the experimental results of all compared methods. We show the real annotations on the clear images to make the targets easier to be found. The detection results of all methods are presented on the raw level-0 infrared remote sensing images. It is worth noting that while YOLO works well for natural images containing objects of various sizes, it may not perform as well on remote sensing images. The results illustrate that some of the improved methods work better for the ISD dataset. SME-Net, which was designed for small objects, mainly focuses on the target scale problem but did not dealt with the environmental impact and noise interference, so it also produced more false alarms and missed targets. For CC-Net, a method with a chained cascade

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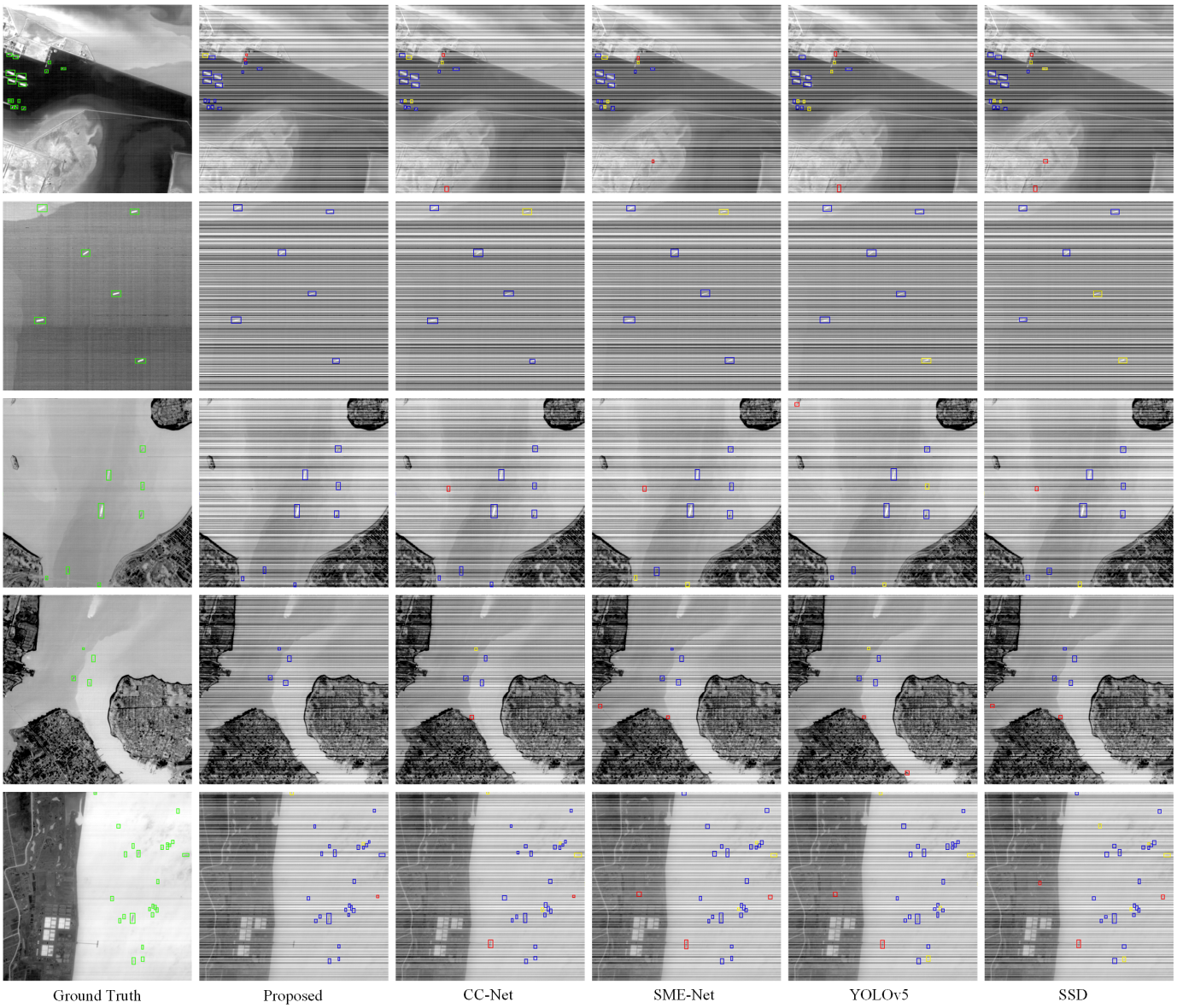

Fig. 11. Prediction results of each method. Green rectangles: ground-truth annotations. Blue rectangles: truth-positive samples. Red rectangles: false alarms. Yellow rectangles: false-positive samples.

structure achieved relatively good results by processing both easy and difficult samples step by step. Furthermore, we can see that the proposed method outperforms other compared approaches in terms of detection performance, as illustrated by the low number of false-positive targets and false alarms.

D. Discussion

In this study, we proposed a progressive task-based universal network for ship detection on raw infrared remote sensing images, which is a dual-supervised learning framework capable of dealing with different tasks simultaneously. Importantly, it is not just a framework that cascading different components but making them serve for the same goal through an effective regulatory mechanism. This distinguishing scheme sets the proposed network apart from other multisupervised frameworks. Besides, in contrast to earlier detection methods that aim to process level-2 image product, the proposed approach can directly process raw infrared images, which is more in line with the real-time requirements of practical applications. We can find from the experiments that the method developed in this article had significant benefits for ship detection from raw infrared images. Admittedly, although the proposed method reduces the application time through an integrated framework, it is not yet lightweight enough for certain applications. Therefore, future work will focus on designing a lightweight network model and exploring the hardware transplantation options for embedded devices, e.g., field-programmable gate arrays (FPGAs) and digital signal processor (DSPs).

V. Conclusion

A well-designed progressive task-based universal framework for raw infrared remote sensing imagery ship detection was proposed in this work. In response to the fact that most previous methods have been studied based on level-2 images

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obtained by high time-consumption processing such as destriping, we proposed a method for ship detection on raw infrared images. The reason why the proposed dual-supervised model can achieve better performance can be attributed to several factors. First, the dual-supervised framework decomposes a difficult problem into different subproblems that are easier to solve. Second, the integrated network framework shortens the processing time of the model. Third, the SDC and ODC components with parallel subnetworks are able to effectively reduce the negative impact of destriping and the false alarm rate of object detection, respectively. The experiments were performed on a set of infrared images of GF-1 satellite, and the results show the superiority of the proposed approach. Although there is still room to improve our approach, we hope that our work offers a fresh and promising ship detection framework and inspires a new direction in the ship detection field.

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